

P-192 weak-object study: singular validation and bounded CM relations

crypto research harness

2026-10-06

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Status and verdict

Evidence cutoff: 2026-10-06. Both native producer runs and the independent review are complete; the review verdict is BREAKS, with separate subclaim dispositions. The validator independently reproduced the singular lane’s fixed public synthetic scalar recovery, so the tested unchecked interface is IMPLEMENTATION_WEAK, not weak P-192. In the valid CM lane the order facts, enumeration counts, theorem bound, and all 103 emitted scalar relations were reproduced. However, the frozen protocol did not specify the canonical-set digest serialization, so exact set identity was not

independently bindable. Explicit maps, HOT/COLD calibration, and charged payoff also remain unavailable. The complete-set negative and full CM experiment are therefore INDETERMINATE; no P-192 structural weakness or speedup was found.

The work uses one fixed, public, synthetic scalar. It does not target deployed keys, live services, or private data. Its purpose is to distinguish an implementation-validation failure from a property of a valid elliptic curve.

Two hypotheses are deliberately kept separate:

1. **Singular implementation-validation lane.** A nodal cubic shares the P-192 field and coefficient $a=-3$, but it is not an elliptic curve and is not isogenous to P-192. The experiment asks whether an unchecked low-level multiplication surface can turn x-only confirmation tags into recovery of the fixed synthetic scalar. The strongest admissible verdict is IMPLEMENTATION_WEAK.
2. **Valid CM/isogeny-class lane.** An exact, bounded class-relation census is run in the endomorphism order attached to the valid P-192 isogeny class. A form relation is not a weakness: global orientation, explicit maps, subgroup transport, point replay, and fully charged payoff must all pass. The preregistered negative verdict is only NO_WEAKNESS_FOUND_WITHIN_SCOPE for the frozen box.

Evidence labels

Statements in this report use these labels:

- **FROZEN:** fixed by approved protocol before execution.
- **DERIVED:** algebraically derived from stated inputs; not a run measurement.
- **INDEPENDENTLY VERIFIED — PRIOR:** replayed evidence from an earlier committed/archived run.
- **PRODUCER-MEASURED:** hash-bound native producer output that was not itself independently rederived.
- **VALIDATED:** independently rederived against the content snapshot. This label applies to the singular scientific joints and named CM subclaims only; it does not override the review's overall BREAKS verdict.
- **OPEN:** required evidence or capability is explicitly unavailable.

Exact object identities

Valid elliptic curve

The valid object is P-192, registered as `icv1-fp192-t31607402316713927207482677199-52e4af59`.

- **Field prime p (FROZEN):**
6277101735386680763835789423207666416083908700390324961279.
- **Short-Weierstrass coefficients (FROZEN):** $a=p-3$; exact b :
 $b=2455155546008943817740293915197451784769108058161191238065$.
- **Prime group order n (FROZEN):**
6277101735386680763835789423176059013767194773182842284081.
- **Trace $t=p+1-n$ (DERIVED):** 31607402316713927207482677199.
- **Compact representation:** EC1P192Cp192h5531c4a08bdb.
- **Full representation UID:** prefix `urn:ec-record:1:sha256:` followed by
5531c4a08bdb64b6e86a6e30e9a08aa57edef7af15ac5f6d4d2a83a53bf2f646.

The exact registry source is docs/curves/registry.json at base revision:

0dc6b7f4a03253e7c7641f99700f7f423f8d07e0.

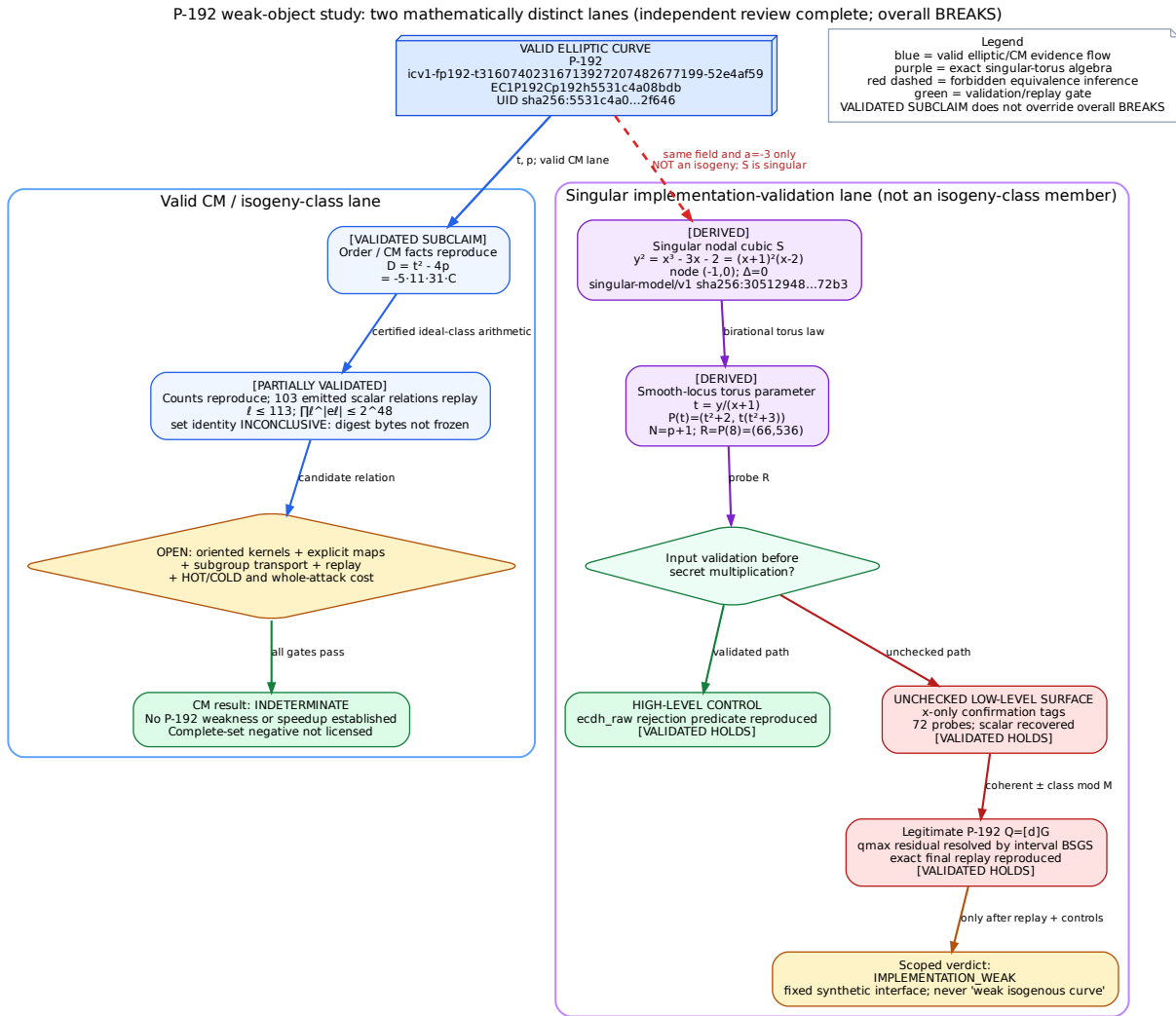


Figure 1: Two lanes with separate identity and evidence gates. Blue is the valid elliptic/CM lane; purple/red is the singular implementation-validation lane.

Its source SHA-256 is recorded in the artifact index.

Singular companion: a typed non-curve object

DERIVED. Define over the same field

$$S : y^2 = x^3 - 3x - 2 = (x + 1)^2(x - 2).$$

For a short-Weierstrass model, the discriminant is $-16(4a^3 + 27b^2)$. At $a=-3, b=-2$, the inner term is $4(-27) + 27(4) = 0$; therefore S is singular, with node $(-1, 0)$. It is not assigned ICV1, EC1, or a curve UID. The exact non-curve preimage is stored in [singular-model.canonical.json](#) with type singular-model/v1; its SHA-256 is 305129485b8a281f5cd73a7a0ec4d82a7f84f432847c1b9d5eb1734983db72b3.

The probe $R=(66, 536)$ satisfies $536^2 = 66^3 - 3 \cdot 66 - 2 = 287296$, is away from the node, and is not on the valid P-192 model because the valid b differs. Sharing p and a does not define an isogeny, an isomorphism, or a curve-family member.

Experiment A — native singular-companion recovery

Protocol: EXP-SCURVE-29040c, approved by DEC-20261006-af8214. Exact inspected protocol SHA-256: ae25388412acb9fbc8a0387f8d4782fe215a493b6bccb6f1f0fa49f8f1b3fde6. Protocol v2 amendment SHA-256 is c5be1d42be520cbadec68c76928f579559f8e1ebfb7fa9aa93c58fa0853512a0; the x-only encoding correction is CORR-20261006-500337. Producer run RUN-SCURVE-af5caf used code commit 6d9d1ed7674cead991549d2060de3e91be1d9a3c and is bound by the content snapshot described in [ARTIFACT_INDEX.md](#).

Frozen algebra and attack path

On the smooth locus set $t=y/(x+1)$. The inverse parameterization and group law used by the independent arithmetic control are

$$P(t) = (t^2 + 2, t(t^2 + 3)), \quad t_1 \star t_2 = \frac{t_1 t_2 - 3}{t_1 + t_2}.$$

The identity is $t=\infty$, inverse is $t \rightarrow -t$, and $R=P(8)=(66, 536)$. Because $p \bmod 12 = 11$ and -3 is a nonsquare, this is the nonsplit nodal torus with order

$$N = p + 1 = 2^{64} \cdot 3 \cdot 5 \cdot 17 \cdot 257 \cdot 641 \cdot 65537 \cdot 274177 \cdot 6700417 \cdot 67280421310721.$$

The frozen point-order certificate must prove $[N]R=0$ and $[N/q]R \neq 0$ for every distinct prime factor q . It may not be inferred from the factorization alone.

The largest factor and its cofactor are

$$\begin{aligned} q_{\max} &= 67280421310721 & (\log_2 \approx 45.93525) \\ M &= 93297598515282145091939986791922449178951680 \end{aligned}$$

The x-only oracle is sign-invariant. Nested composite-order probes therefore retain one coherent pair $\{r, -r \bmod m\}$ rather than independently choosing a sign for every factor. The frozen sequence contains 64 lifts by two and eight odd factors, so the expected primary-oracle count is exactly 72. Candidate comparisons are counted separately and must not be relabelled as oracle calls.

After recovery modulo M , the legitimate public key $Q=[d]G$ on P-192 is required to solve the remaining interval. One shared negation-BSGS table is frozen at `table_bits=24, m=5,800,019, stride 11,600,037`, no more than 5,800,018 finite baby entries, and no more than 5,800,018 giant positions per orientation. Every fingerprint hit requires exact replay.

Fixed public synthetic fixture

- **Seed:** P192-SINGULAR-V1; hex 503139322d53494e47554c41522d5631.
- **Scalar mapping:** $1 + \text{OS2IP_BE}(\text{SHA256}(\text{domain} \parallel \text{seed})) \bmod (n-1)$.
- **Expected d in hex:**
8b70f85429c81223e88be1a60590627a3c8e558a24dd5180.
- **Expected d in decimal:**
3419090462492416794041408622244871729251768983841006244224.
- **Unordered residues modulo M:**
 - 78222222987683236539393919745863586265518464.
 - 15075375527598908552546067046058862913433216.
- **True residual k:** 36647143301682. Check only after recovery; never use it to prune the search.

Success, falsification, and controls

An IMPLEMENTATION_WEAK result requires all of the following, not merely successful torus arithmetic:

- the exact P-192 and singular-point/order certificates pass;
- the actual unchecked scalar_mul_secret surface is used for every primary query and exposes only the frozen x-derived HMAC tag;
- one coherent plus/minus class is recovered after exactly 72 primary calls;
- the legitimate P-192 residual solve returns the fixed d;
- independent final replay proves $[d]G=Q$;
- the safe high-level ecdh_raw path rejects every singular probe before multiplication; and
- exhaustive $p=59$, wrong-parameter, mutated-transcript, full-point positive, and independent-torus controls all give their preregistered verdicts.

If instrument controls pass but the x-only transcript cannot recover the coherent class, or neither residual orientation replays to Q, the hypothesis is falsified. A timeout, OOM, or missing receipt is INDETERMINATE, not evidence that the valid curve is strong.

Producer result and independent validation

VALIDATED AT THE STATED INTERFACE; OVERALL REVIEW BREAKS ON ARCHIVE METADATA.

The native run recovered the exact fixed scalar listed in the fixture above; a source-blind checker independently reproduced all 72 tags, 1,245,111 candidate comparisons, the coherent global sign, the BSGS equation, the residual k, the recovered d, and $[d]G=Q$. Order, torus, transcript, mutation, $p=59$, safe-path rejection, and full-point-model controls passed. Both singular scientific joints are HOLDS. No speedup is asserted.

- **Valid P-192 reference:** approximately $\sqrt{n}$ group work; a matched native reference was not run.
- **Nested residues:** 72 primary x-only queries and 1,245,111 candidate comparisons, kept as different counted units.
- **Shifted-negation-v2 BSGS:** 5,800,018 baby entries. The exact hit has orientation -1, giant index 3,159,226, baby index 989,698, and sign -1.
- **Whole pipeline:** success; 20.809205507 seconds wall time, 133,616 KiB maximum RSS, and every frozen control passed.

Correctness is certificate-backed at the producer layer. A ratio to a matched reference remains unavailable and is not meaningful until every phase is expressed in the same counted unit.

Wall time is secondary engineering evidence. A performance claim additionally requires the repository's isolated A/A and interleaved paired-measurement protocol; this single recovery run is a deterministic feasibility result, not a timing confidence interval or prevalence estimate. The supported

object is the exposed implementation interface. The singular cubic remains neither an elliptic curve nor a P-192 isogenous representative.

The original snapshot receipt misspelled the singular producer-origin commit. All 62 content hashes and both manifest chains passed, but strict provenance therefore broke. The immutable receipt remains unchanged; additive correction CORR-20261006-cfc18a overlays the actual commit 8b81f8efa613766a76eaf46465892f22b37fad5d. This administrative defect does not erase the independently reproduced arithmetic, and the correction does not turn the overall review verdict into a pass.

Experiment B — exact weighted CM class-relation census

Protocol: EXP-SCURVE-647ade, approved by DEC-20261006-af8214. Exact inspected protocol SHA-256: 0fe122a0c6be3e2118b4d9c276a1085285c7e6a2cee38d15bbeed8099fa17145. Protocol v2 amendment SHA-256 is 029fcf0ed147af8eb590e770bed9c2a023eadfd3bca8567d0297caa6130d3a33. Producer run RUN-SCURVE-81c5e4 used corrected code commit 78ba2da296c505621feb0b05c11507f3b5ae6669 and is bound by the content snapshot described in [ARTIFACT_INDEX.md](#).

Frozen order and generator set

PRODUCER-MEASURED; certificate emitted. The Frobenius discriminant is

$$D = t^2 - 4p = -24109379060336110122544161233113975664949272517896865359515 = -5 \cdot 11 \cdot 31 \cdot C,$$

where $C=14140398275856956083603613626459809774163796198179979683$. The run emitted and replayed a recursive Pocklington certificate for C , then certified that D is squarefree and fundamental, the Frobenius conductor is one, and $\text{End}(E)=\mathbb{Z}[\pi]=0_D$. The native p/n checks remain fixed-base probable-prime checks with pinned SEC 2/FIPS provenance, not newly constructed recursive proofs.

The frozen ramified roots are 5: [2], 11: [3], and 31: [7]. The frozen split roots are:

ell	roots of $X^2 - tX + p \bmod \text{ell}$
13	2, 5
23	21, 22
37	12, 35
43	8, 26
73	60, 67
89	6, 83
101	17, 70
103	5, 36
107	56, 68
113	26, 42

Each root and orientation must be recomputed. A positive split generator uses the smaller root; global conjugates are canonicalized by requiring the first nonzero split exponent to be positive. Ramified exponents are unsigned.

Exact boundary and theorem control

The census includes every canonical exponent vector with $L = \text{product } \text{ell}^{|\mathbf{e}_{\text{ell}}|} \leq 2^{48}$; empirical weights may rank but may not prune. The preregistered cardinalities are 9,948,061 oriented vectors, 4,974,348 canonical vectors including zero, and 4,974,347 canonical nonzero vectors, under a hard cap of 5,000,000 canonical states. A differing count invalidates the run.

For a non-scalar $\alpha = u + v\pi$, the protocol must independently certify

$$N(\alpha) \geq \left\lceil \frac{|D|}{4} \right\rceil = 6027344765084027530636040308278493916237318129474216339879 \approx 2^{191.941}.$$

If all premises pass, the exact 2^{48} lane can contain only scalar ramified relations. This is a proof-controlled scope statement, not a claim about all relations or all attack mechanisms.

CM degree frontier: exact scope versus non-scalar theorem bound

Unit: $\log_2$ ideal norm / geometric degree. Derived and preregistered values only; no run or timing result is plotted.

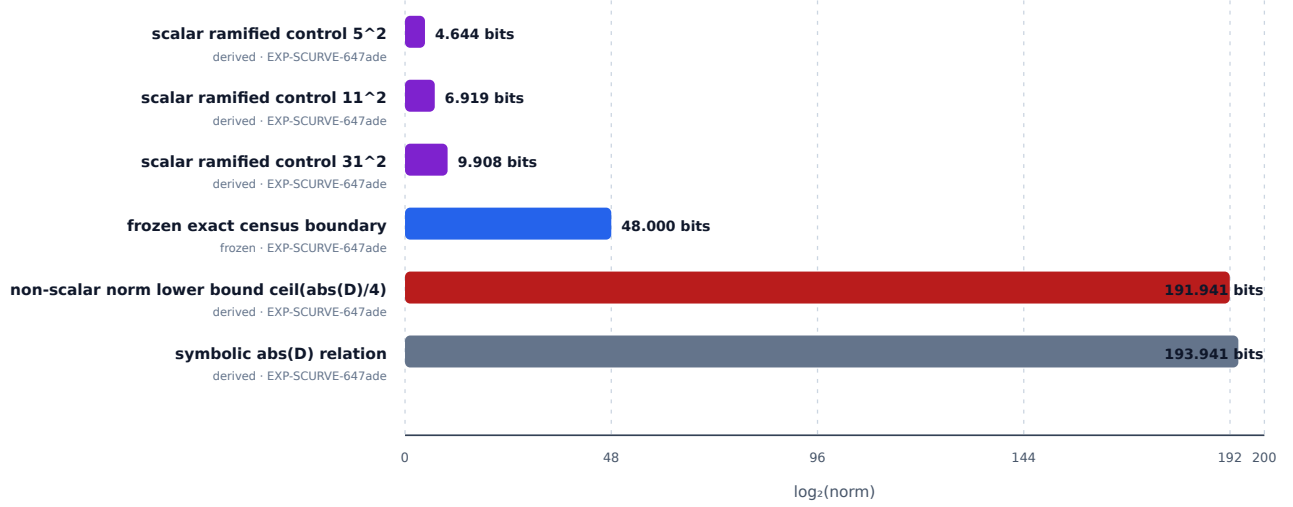


Figure 2: Derived CM degree/norm values. The plot contains no empirical run or timing data.

The controls are $g5^2$, $g11^2$, and $g31^2$ on P-192; the positive non-scalar toy relation $g2^3=1$ at $D=-23$; and the symbolic P-192 relation $g5 \cdot g11 \cdot g31 \cdot gC=1$ with $\alpha=\sqrt{D}=2\pi-t$. The last relation is genuine algebraically but is outside the buildable $\text{ell} \leq 113$ set: a C-degree edge is unavailable, and its action on $E(F_p)[n]$ does not by itself improve a DLP. It must remain a rejected symbolic control.

Result and payoff gates

PRODUCER-MEASURED. Complete exact algebra census; no non-scalar relation in the frozen box. Both enumeration orders returned 9,948,061 oriented vectors, 635 conjugation-fixed vectors, 4,974,348 canonical vectors including zero, and 4,974,347 canonical nonzero vectors, with identical set digests. The exact $L=2^{48}$ layer is empty with the canonical empty-input SHA-256. There are 104 scalar-principal states including zero, hence 103 emitted nonzero relations; all are scalar ramified, all 103 HNF/form replays passed, and zero are non-scalar.

INDEPENDENT REVIEW. The order/discriminant/conductor facts, generator roots, theorem lower bound, all enumeration cardinalities, all 103 emitted relation certificates, and the mutation controls were independently reproduced. The 103 emitted relations are therefore validated as scalar. The review could not reproduce the producer's canonical-set digest from the frozen protocol, because the protocol names a digest but never defines its domain separator, vector serialization, XOR/sum chunking, or byte order. The producer source uses a domain-separated construction, but learning that after blind sealing cannot retroactively repair the protocol. Thus matching counts do not independently certify exact set identity.

- **Order/CM proof — pass:** pinned tuple, Hasse uniqueness, recursive C proof, fundamental D, and conductor one.

- **Exact membership — inconclusive:** forward/reverse producer counts and digests agree, and counts independently reproduce, but the digest encoding was not frozen and exact set identity is not independently bound.
- **Algebraic replay — validated:** 103 scalar-ramified records accepted and zero rejected.
- **Global map realization — OPEN:** no non-scalar candidate exists, and globally oriented explicit maps and terminal isomorphisms are not implemented.
- **HOT/COLD accounting — OPEN:** zero of 368 required calibration records; missing tuples are null, not zero.
- **Whole payoff — not passed:** no charged attack or canonicalization inequality exists.

The exact search took 28.239448942 seconds and the full verifier replay took 11.462448644 seconds; aggregate phase wall time was 39.733819654 seconds with 639,024 KiB peak RSS. These are single-run engineering observations, not a speed comparison. The degree-linear proxy was not substituted for missing HOT/COLD operation costs.

Two preserved attempts found defects in the research harness rather than in P-192: an incorrect Bézout coefficient plus duplicated factor in generic quadratic-form composition, and a one-ULP JSON float round-trip mismatch that the exact verifier correctly rejected. Commits bcd59e1 and 78ba2da repaired them and added full-census, one-ULP, and NaN regressions before attempt 3.

The producer’s bounded negative remains a useful observation, but independent review does not license NO_WEAKNESS_FOUND_WITHIN_SCOPE for the complete set. Because exact set identity, map/calibration/payoff evidence, and map-specific controls are unavailable, the CM experiment is INDETERMINATE, not evidence that P-192 is globally strong or weak. No non-scalar relation was found in the producer census, and none appears among the independently replayed emissions.

What will count as “weak”

The word *weak* is reserved for an auditable statement with an exact object, threat model, boundary, and verified advantage.

- **INVALID_INSTANCE:** a purported elliptic-curve candidate is singular, malformed, or otherwise fails identity preconditions. The singular companion has this object-level classification even though it remains useful for an implementation-validation test.
- **IMPLEMENTATION_WEAK:** the actual unchecked surface, fixed x-only oracle, end-to-end planted-key recovery, independent replay, and every control pass. The singular cubic is still not a weak elliptic or isogenous curve.
- **CLASS_WEAK:** a certified attack applies across the declared valid isogeny class, with subgroup transport, replay, and material advantage.
- **WEAK_REPRESENTATIVE_EXISTS:** at least one valid class representative has a certified weakness, without claiming every representative shares it.
- **SOURCE_TRANSFER_WEAK:** a certified map transfers a concrete weakness back to the named source curve with all conversion costs charged.
- **NO_WEAKNESS_FOUND_WITHIN_SCOPE:** the frozen CM box is exhausted with exact counts, independently bindable set identity, proofs, and replay, and no candidate passes the buildability and payoff gates. This gate was not met in the present CM review; it is a criterion, not the result.
- **INDETERMINATE:** mandatory map, cost, control, or replay evidence is unavailable. Resource exhaustion or unsupported tooling is not strength evidence.

The safe public ECDH path’s rejection of every singular probe is recorded as a passing control, not as a separate final verdict.

P-192’s roughly 96-bit generic security level and legacy status are baseline properties, not discoveries of this campaign.

Confidence model

Deterministic claims

Primality, factorization use, point order, class-group products, HNF identities, enumeration membership, map equations, and final scalar replay are deterministic claims. Statistical confidence is the wrong tool for them. Confidence comes from self-contained certificates, exact counts/digests, a second enumeration order, mutation-negative controls, and an independent verifier that does not trust the producer's intermediate state.

The CM producer artifacts support a finite-census observation and independently reproduced counts, but the review does not support the stronger complete-set statement because the digest encoding was not frozen. No confidence interval can repair an underspecified deterministic identity check; a new additive protocol must define the bytes and rerun or re-digest them independently.

Timing and implementation variability

Any wall-clock comparison must first measure A/A spread, then interleave baseline and candidate for at least five paired rounds under the repository's isolated runner. Report counted native operations as primary, plus median, minimum, contention flags, and a 95% paired confidence interval. A runtime improvement is supported only if the interval excludes no improvement and all outputs/digests match. The recorded 20.809-second recovery and 39.734-second CM pipeline are single executions, so no timing comparison or confidence interval is asserted.

Generalizing beyond one fixture

One fixed synthetic scalar can demonstrate a mechanism and exact recovery; it cannot estimate a prevalence rate across keys, implementations, or curves. A later population claim would require a preregistered sampling frame, independent holdouts, multiple keys per implementation, explicit failure retention, exact binomial intervals, and multiplicity control across curve/candidate searches. No such population inference is made here.

Prior evidence and non-duplication

The prior committed P-192 isogeny walk [research/isogeny_walk_p192_20261004/README.md](https://github.com/autoresearcher/isogeny-walk/p192-20261004/README.md) is **INDEPENDENTLY VERIFIED — PRIOR**, not a result of these experiments. Its 20,000-curve run is p192-71c04205135fcf8f, archived under s3://crypto-autoresearcher/isogeny-walk/runs/p192-71c04205135fcf8f. It reported 20,000 valid curve records, 128,700 edges, zero failures, and replay pass, but explicitly established no DLP weakness or per-curve IC difference. The new exact CM census asks a different, relation-focused question and must not rewrite that frozen evidence.

The repository's typed-link rules ([docs/curves/ic/curve-links/README.md](https://github.com/autoresearcher/isogeny-walk/docs/curves/ic/curve-links/README.md); source hash in the artifact index) require exact endpoints, kernels/maps, dual status, subgroup transport, and replay before an isogeny route can carry a DLP claim. That rule is the basis of the CM payoff gate and the red "NOT an isogeny" separation in the diagram.

Open obligations

- Independently accept or reject additive custody correction CORR-20261006-cfc18a; do not edit the immutable snapshot receipt or erase the historical review break.
- Freeze the CM canonical-vector byte encoding, domain separator, digest aggregation, and byte order in a prospective amendment, then rerun or independently re-digest the complete set.
- Keep explicit global maps, terminal isomorphisms, HOT/COLD calibration, and fully charged payoff open. Do not fabricate them from degree or wall time.
- Preserve separate evidence records and the post-result revise decision; never merge the singular implementation conclusion into the valid CM/isogeny conclusion.

- Regenerate and visually inspect this report PDF after the final archive hashes are incorporated.
- Do not extend the conclusion to sect113r1, other standardized curves, other APIs, or a population of implementations without separately frozen inputs and evidence.

Sources and provenance

Repository and protocol sources

1. P-192 registry entry: [docs/curves/registry.json](#), exact EC1/UID above; source hash and revision recorded in the artifact index.
2. Repository curve parameters: [src/ecc/curve_zoo.rs](#), source hash recorded in the artifact index.
3. Prior P-192 walk: [research/isogeny_walk_p192_20261004/README.md](#), with report, run, and store hashes in the artifact index.
4. Frozen experiment records EXP-SCURVE-29040c and EXP-SCURVE-647ade, their v2 amendments, archive correction CORR-20261006-9ef467, and the additive origin overlay CORR-20261006-cfc18a, approved or recorded under the campaign ledger; exact hashes are recorded above and in the artifact index.
5. Independent review TASK-20261006-7d34e7, evidence records EV-SCURVE-44f051 and EV-SCURVE-8e5c70, and post-result decision DEC-20261006-581065; their exact hashes and archive commit are recorded in the artifact index.
6. Evidence and accounting rules: [AGENTS.md](#), SHA-256 recorded in the artifact index.
7. The canonical audit-curve skill and KN-TECH-6a2ef9 weak-curve audit synthesis in crypto-autoresearcher; these define the verdict taxonomy and keep invalid-input, representative, class-wide, transfer, and implementation claims separate.

External parameter and terminology sources

- NIST, [FIPS 186-4](#), archived standard containing the legacy NIST prime-curve parameters.
- SECG, [SEC 2 version 2.0](#), secp192r1 domain parameters.
- Andrew V. Sutherland, “[Isogeny Volcanoes](#)”, for isogeny-volcano terminology. The report does not treat terminology as a certificate for any new edge.

The complete artifact set and remaining technical obligations are in [ARTIFACT_INDEX.md](#).